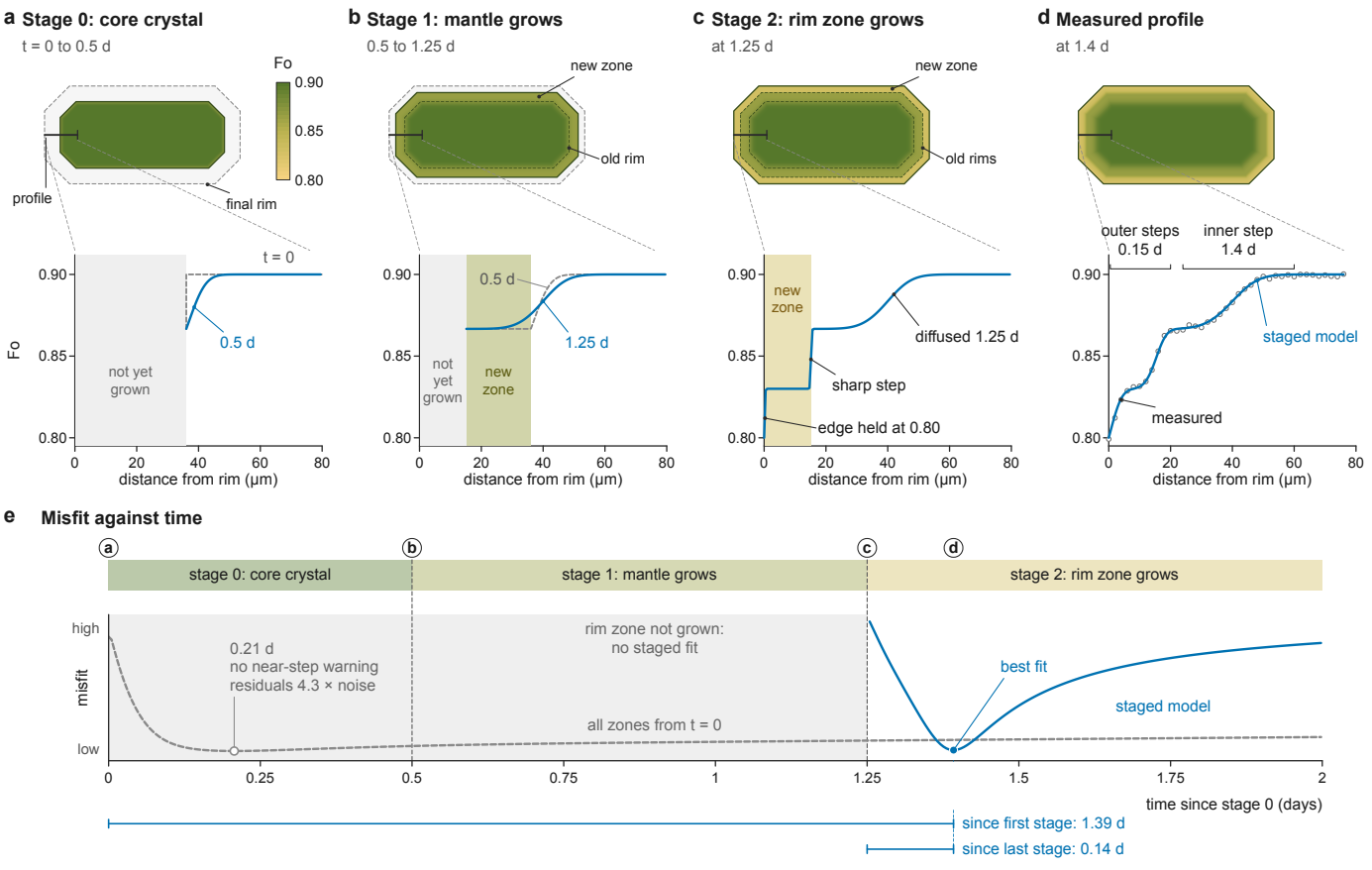

a Stage 0: core crystal
t = 0 to 0.5 d
Fo
0.90
0.85
0.80
profile
final rim
t = 0
0.5 d
not yet grown
Fo
distance from rim (μm)
b Stage 1: mantle grows
0.5 to 1.25 d
new zone
old rim
0.5 d
1.25 d
not yet grown
new zone
c Stage 2: rim zone grows
at 1.25 d
new zone
old rims
new zone
diffused 1.25 d
sharp step
edge held at 0.80
d Measured profile
at 1.4 d
outer steps 0.15 d
inner step 1.4 d
staged model
measured
e Misfit against time
a
b
c
d
stage 0: core crystal
stage 1: mantle grows
stage 2: rim zone grows
high
misfit
low
0.21 d
no near-step warning
residuals 4.3 × noise
rim zone not grown:
no staged fit
all zones from t = 0
best fit
staged model
time since stage 0 (days)
since first stage: 1.39 d
since last stage: 0.14 d